

CONTACTS

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EDUCATION

University of Texas at Austin, Austin, TX, USA

PhD in Mechanical Engineering, Expected to Graduate in 2027

Northwestern University, Evanston, IL, USA

Master of Science in Robotics, December 2021

Saint Louis University, St. Louis, MO, USA

Bachelor of Science in Mechanical Engineering, May 2020

WORK

Robot Mechatronics Intern

Agility Robotics | Salem, OR | 06/2025 - 09/2025

- Built a **tendon-driven finger** prototype for Digit and a test bed for evaluating cable transmission properties, including tension propagation and phase lag under cyclic loading.
- Developed a frequency-domain **system identification** framework to characterize cable transmissibility, linearity, and dynamic stiffness, establishing design metrics for **cable-driven mechanisms**.

Graduate Research Assistant

UT Austin & Sony Group Corporation | Austin, TX & Tokyo, Japan | 08/2023 - Present

- Designed and fabricated **dexterous robotic hands** with anthropomorphic fingertips for sub-millimeter precision in small-part manipulation.
- Developed a **whole-body teleoperation** framework for a 7-DOF manipulator and an 8-DOF hand, utilizing **hybrid position/force control** to manage high-DOF multi-kinematic-chain redundancy.
- Architected a **shared-control** scheme for dexterous **in-hand manipulation** with **tactile sensors**, stabilizing multi-contact interaction while reducing computational overhead and teleoperator effort.

Research Intern

HQ Tech | Daejeon, South Korea | 05/2017 - 08/2017

- Developed vision-based quadcopter **UAV control** for reservoir flow measurement.

RESEARCH

Morphological Design of Contact Interfaces for Dexterous Manipulation

Research direction: designing contact-interface morphology to shape local contact mechanics, tactile sensitivity, and controllability in dexterous manipulation.

- Rigid-soft fingertip design with fingernails and compliant pulp for edge manipulation, grasp stability, and force-transparent interaction.
- Contact-mechanics-informed design principles linking fingertip morphology, tactile sensitivity, and dexterous manipulation performance.
- Shared-control and learning-based control strategies for stabilizing multi-contact in-hand manipulation.

Humanoid Upper-Body Design for Whole-Body Manipulation

Research direction: designing humanoid wrists, forearms, arms, and upper-body structures as sensing-rich, mechanically intelligent platforms for whole-body manipulation.

- Wrist and forearm mechanism design with optimized joint axes, compact transmission, and contact-aware morphology.
- Mechanism optimization and control co-design to reduce joint reaction forces, actuator effort, and unsafe contact loads.
- Whole-body sensing-enabled upper-body architectures for physical transparency, teleoperation, and contact-rich manipulation.

PUBLICATIONS	S. Dalla Gasperina [†] , DH. Kang [†] , H. Zhang [†] , A. Galvan [†] , J. D. Ramirez, A. Kim, M. Helwig, K. Yokoyama, T. Ueno, T. Narita, A. Majewicz Fey, A. D. Deshpande, and L. Sentis, “A Bilateral Teleoperation Framework for Dexterous Manipulation,” under review, 2026.	
	[†] <i>Equal contribution.</i>	
	DH. Kang , A. Kim, M. Seo, K. Yokoyama, T. Narita, and L. Sentis, “PLATO Hand: Shaping Contact Behavior with Fingernails for Precise Manipulation,” <i>IEEE Robotics and Automation Letters</i> , vol. 11, no. 7, pp. 8648–8655, 2026. doi: 10.1109/LRA.2026.3699243	
	A. Kim, DH. Kang , M. Helwig, M. Seo, K. Yokoyama, T. Narita, and L. Sentis, “ARISTO Hand: Sensing-Driven Distal Hyperextension for Fine-Grained Manipulation,” <i>2026 IEEE/ASME International Conference on Advanced Intelligent Mechatronics (AIM)</i> , 2026.	
	M. Kojouharov, DH. Kang , D. Rowland, R. Mykhailyshyn, L. Sentis, and A. Majewicz Fey, “Soft Vortex Gripper for Dexterous Manipulation using Hand-Like Robots,” <i>IEEE International Conference on Robotics and Automation (ICRA)</i> , 2026.	
TECHNICAL SKILLS	SH. Bang, C. Gonzalez, G. Moore, DH. Kang , M. Seo, R. Gupta, and L. Sentis, “RPC: A Modular Framework for Robot Planning, Control, and Deployment,” <i>2025 IEEE/SICE International Symposium on System Integration (SII)</i> , pp. 1142–1148, 2025. doi: 10.1109/SII59315.2025.10871130.	
	C. Gonzalez, S. Lee, F. Montano, S. Ortega, DH. Kang , M. Jaiswal, J. Jiao, and L. Sentis, “Design of a Person-Carrying Robot for Contact Compliant Navigation,” <i>Proceedings of the ASME 2023 IDETC-CIE</i> , vol. 8, V008T08A094, 2023. doi: 10.1115/DETC2023-117215.	
	Programming: C, C++, Python, MATLAB	
	Mechanical Design: Creo, Onshape, SolidWorks, Ansys	
	Robot Dynamics & Control: mechanism optimization, whole-body control, learning-based control	
TEACHING EXPERIENCE	Simulation: MuJoCo, Drake, Isaac Sim, Isaac Lab	
	Embedded Robotics: FreeRTOS, STM32CubeIDE, CAN, EtherCAT	
	Mechatronics Lab (ME 140L) TA <i>University of Texas at Austin 08/2022 - 12/2022</i>	
	Mechanical Engineering Lab (MENG 3001) TA <i>Saint Louis University 01/2020 - 05/2020</i>	
	Academic Tutor <i>Firm Foundation Tutoring Program 09/2016 - 03/2020</i>	
HONORS AND AWARDS	• Developed course syllabi, study guides, assessments, and supplemental documents for students in grades 4 to 9. • Taught Physical Science, Algebra, Writing Composition, and Reading Literature.	
	Grand Challenges Scholar , National Academy of Engineering, 2020	
	Parks College Innovation Challenge 1st Place , Saint Louis University, 2018	
	Dean’s List , Saint Louis University, 2018	
RELEVANT COURSEWORK	Robotic Manipulation	Embedded Systems in Robotics
	Sensing, Navigation, and ML	Design and Control of Humanoid Robots
LANGUAGE SKILLS	Advanced Mechatronics	Sensory Acquisition
	Brain, Body, and Robotics	Haptics and Teleoperated Systems
LANGUAGE SKILLS	English: Native	
	Korean: Native	